



Inputs

Active Client Layer Updates

candidate updates
from active clients

$$\Delta W_k^{(l)}$$

RawScore from Stage 3

trust-weighted
permission score

$$RawScore_k$$



Control & Decision

Privacy Sensitivity

exposure risk
of layer l

$$S_{priv}^{(l)}$$

Utility Sensitivity

contribution to
global direction

$$S_{util}^{(l)}$$

Security Deviation Sensitivity

deviation from
trusted reference

$$S_{sec}^{(l)}$$

risk fusion

Total Defense Requirement

weighted fusion of three signals

$$S_{total}^{(l)} = w_p S_{priv}^{(l)} + w_u S_{util}^{(l)} + w_s S_{sec}^{(l)}$$

threshold control

Layer Threshold

admission barrier
for this layer

$$\theta^{(l)}$$

radius control

Clip Radius

maximum safe
update magnitude

$$C^{(l)}$$



Constrained Execution



Survivor Selection

keep only qualified clients

$$\Phi^{(l)} = \{k : RawScore_k \geq \theta^{(l)}\}$$

survivor
filtering



Weight Re-normalization

redistribute weights over survivors

$$\sum \tilde{w}_k^{(l)} = 1$$

reweight
survivors



Dynamic L2 Clipping

project updates into safe radius

$$\|\hat{\Delta} \tilde{W}_k^{(l)}\|_2 \leq C^{(l)}$$

clip
magnitude



Secure Layer Aggregation

aggregate filtered clipped survivors

$$\Delta W_{global}^{(l)} = \sum \tilde{w}_k^{(l)} \hat{\Delta} \tilde{W}_k^{(l)}$$

aggregate
securely



Global Layer Update

applied to global model

$$\Delta W_{global}^{(l)}$$